

WissKI Barrels 3D Data Model - Quickstart to FAIR 3D Heritage Documentation

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Abstract

The [WissKI Barrels](#) project, funded by the Deutsche Forschungsgemeinschaft (DFG) under the [VIGO](#) programme (“Jointly Organising Responsibility for Information Infrastructures”) and carried out at the Germanisches Nationalmuseum Nürnberg, develops domain-specific data models for material culture documentation and delivers them as ready-to-use Drupal/[WissKI](#) web applications. Each barrel ships a CIDOC-CRM-based ontology layer, pre-configured entry and display masks, authority connections (GND, Getty, Geonames, Wikidata), and open endpoints (REST, SPARQL), so institutions can begin FAIR-aligned cataloguing without designing a model from scratch. The contribution outlines design rationale, modelling workflow, and first results, and argues that the semantic layer Barrels provides complements 3D publication infrastructures.

Problem. Many heritage institutions digitise objects but lack practically implemented documentation standards that are also interoperable across projects. **CIDOC CRM** offers a generic semantic backbone, yet application-level operationalisation forces a trade-off: overly general designs lose discriminatory power for domain-specific questions; overly local ones block comparison and reuse. Disciplines concerned with material culture, such as art and architectural history, archaeology, provenance research, and conservation, need models that are specific enough to be useful yet open enough to interoperate. Without them, sophisticated 3D viewers and repositories stay semantically shallow: objects display, but actors, events, provenance chains, and interpretive context remain opaque.

Approach. The project resolves that trade-off by synthesising application ontologies and implemented models from more than seventy museum, archive, library, and research-network projects. Domain experts, curators, and developers iterate to distill shared patterns, most notably an event-centred idiom spanning provenance research, conservation documentation, and object biography. At the time of submission the involved actors came from different backgrounds in data management (universities, libraries, museums, working groups, data competence centres, research institutions). Till the end of

the project we try to involve more actors from national and international contexts like NFDI or CIDOC to ensure a broader integration and interoperability.

Technically the Barrels follow five principles: consistent, non-redundant entities; depth matched to scholarly relevance; explicit ambiguity and multiperspectivity; integrated norm data; modular extensibility. The graph always instantiates the full intermediate structure; switchable interface modes (e.g. simplified vs. detailed provenance) tune visible complexity without blocking later enrichment.

Results, assessments, and examples. By the time of the conference, four barrels will be available: **Base** (covering object, actor, place, event, etc.), **Provenance** (documenting assessments of (claimed) ownership, custody, stay, etc.), **Conservation/Restoration** (including samples, modifications, treatments, etc.), and **3D** (focused on 3D models, digitalisation processes, paradata, and the documentation of 3D-specific activities and metadata etc.). Each module is offered as a container, a deployable cloud instance, or for direct installation; all are open-source and can be integrated or migrated into other systems. The model stack is intended for seamless connection to Linked Open Data ecosystems (i.e. through TS4NFDI Terminology Service or Minimal Dataset Recommendations of the DDB), allowing 3D, 2D, and textual assets to be linked and described—thus creating comprehensive metadata and paradata as highlighted by this call. Furthermore, thanks to the collaboration with the DFG 3D Viewer project, 3D Barrel offers an extension possibility of setting up a METS/MODS compliant endpoint for harvesting the 3D-data with the basic metadata to data aggregators, like DDB or further Europeana.

The presentation will include assessments of interface design choices, discuss modifications made to the underlying models, and offer sample scenarios—demonstrating how institutions with existing 3D viewers can substantially enrich their collections with semantically robust, FAIR-compliant catalogues by adopting or customizing a barrel.

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